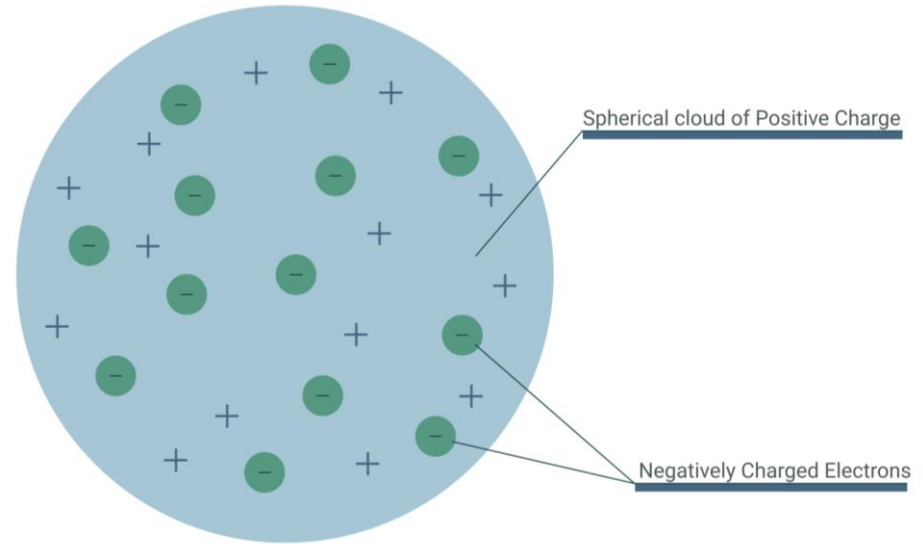


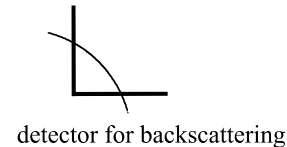
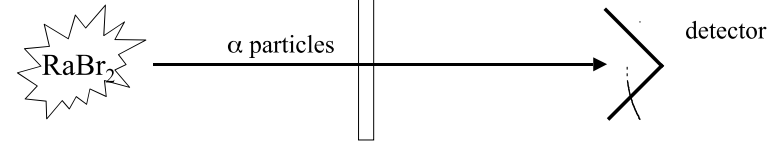
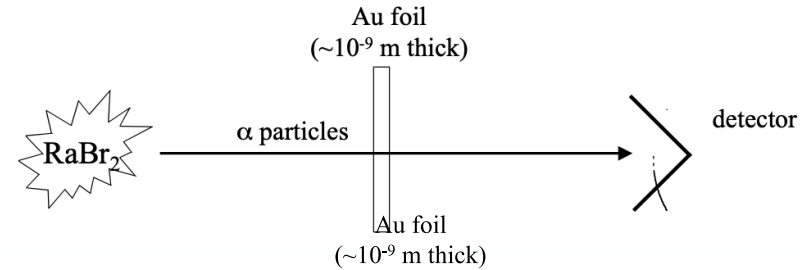
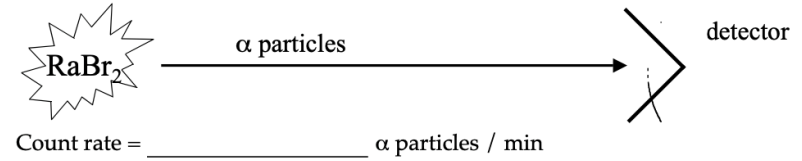
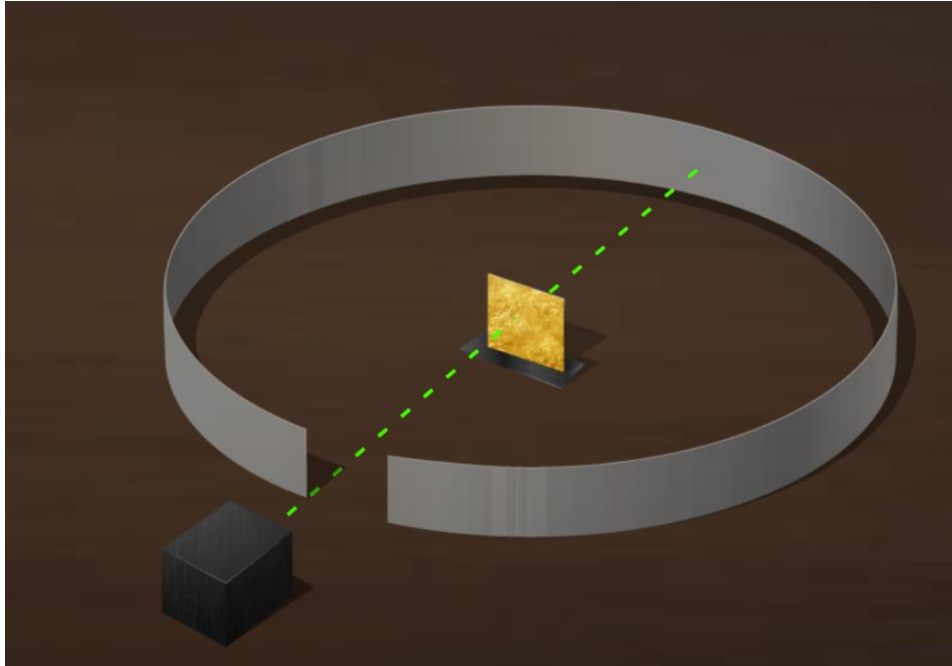
Structure of the Atom

- In 1909, the prevailing theory of atomic structure was the plum pudding model.
- In this model, proposed by J. J. Thomson, the electrons were thought to be floating around in a soup of positive charge.



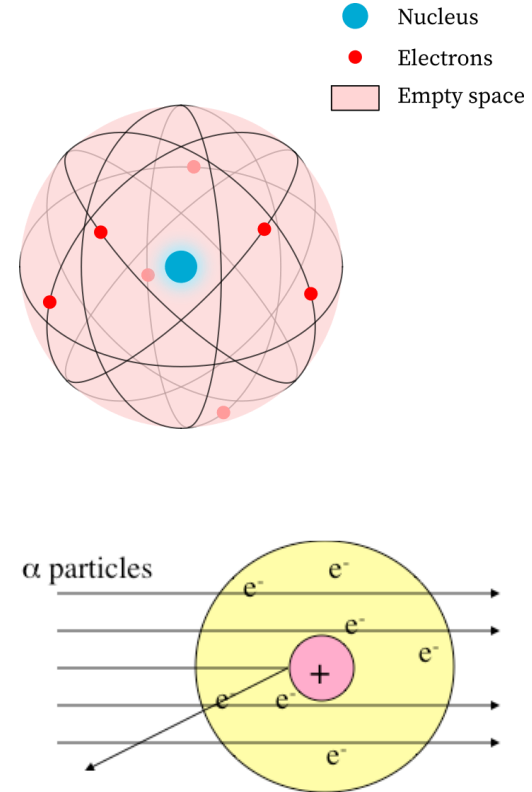
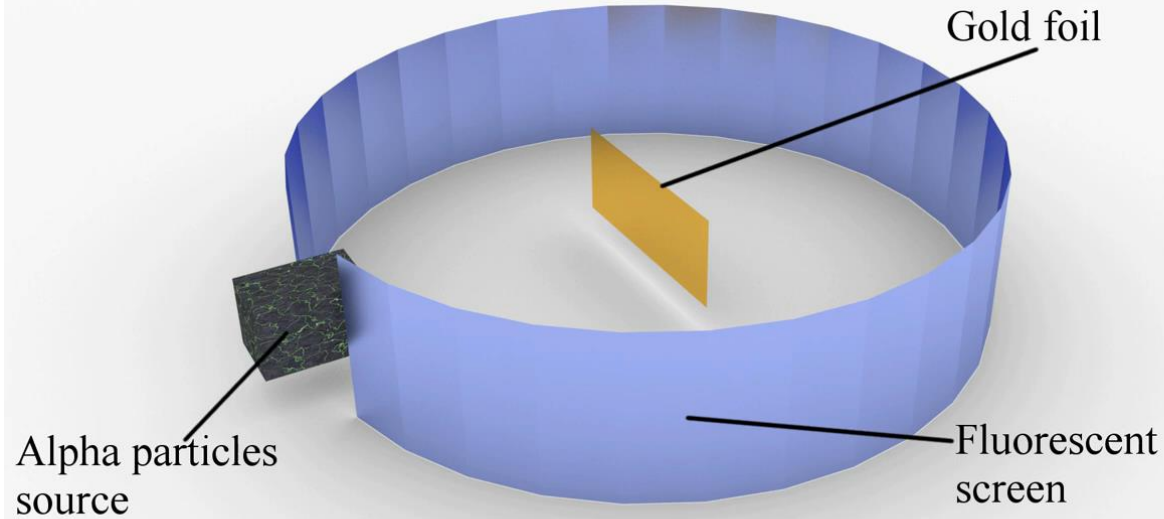
Rutherford's Atomic Model

Ernest Rutherford (Cambridge University, England, 1871-1937) studied α emission from newly-discovered radioactive elements.



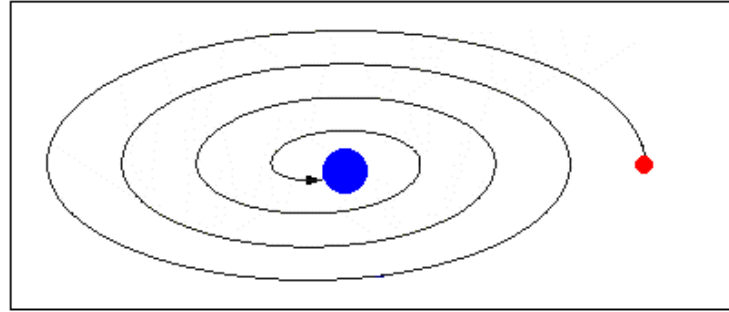
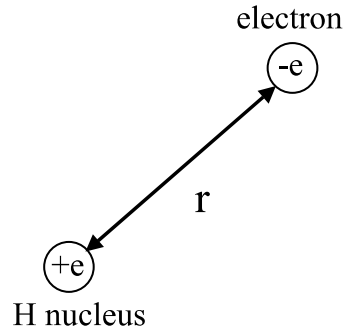
"It was almost as incredible as if you fired a fifteen-inch shell at a piece of tissue paper and it came back and hit you"

The Rutherford Experiment

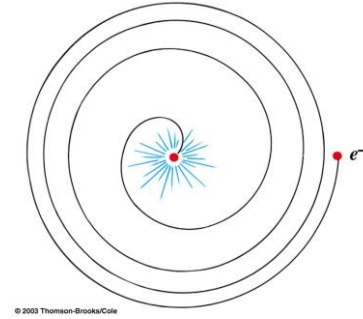


Rutherford's Atomic Model

Drawback



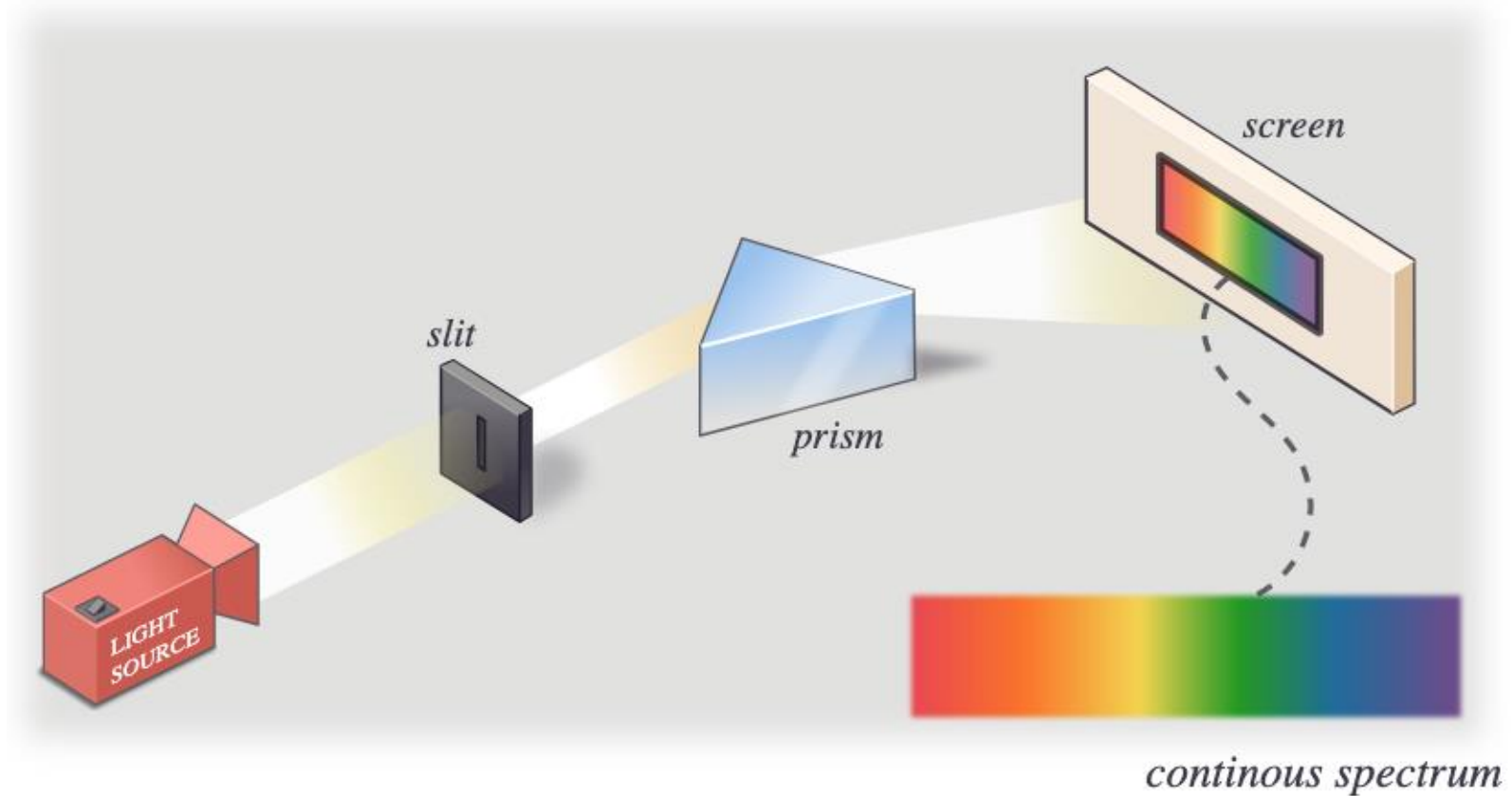
The electron should fall on the nucleus.



The predicted death spiral of the electron!

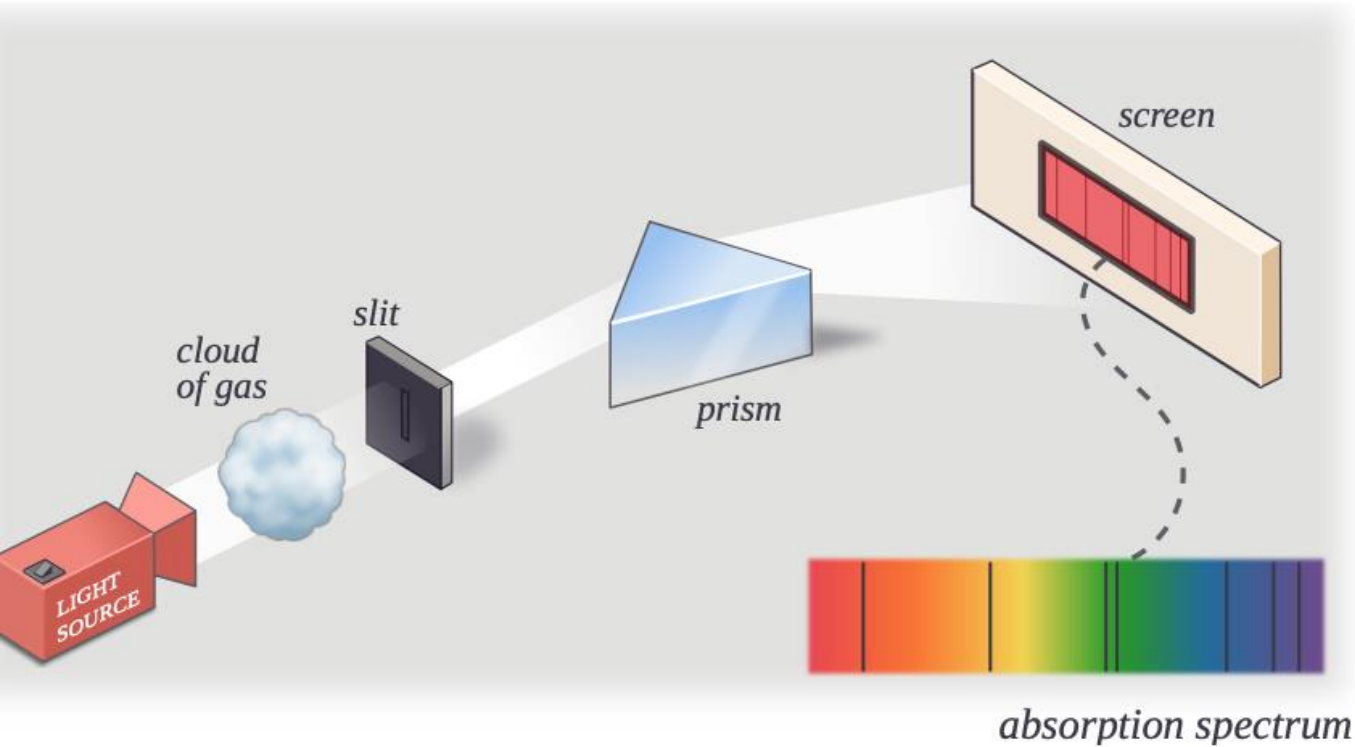
- According to classical theory, atoms in the model for Rutherford's are not stable.
- The motion of moving electrons should cause them to radiate energy in their orbits and to quickly execute a death spiral and collapse into the nucleus.
- The time taken for this collapse is estimated around 10^{-8} s, which is very small.
- If the electrons are not revolving but stationary, still the electrons will fall into the nucleus by the electrostatic force between both.

Hydrogen Atomic Spectrum



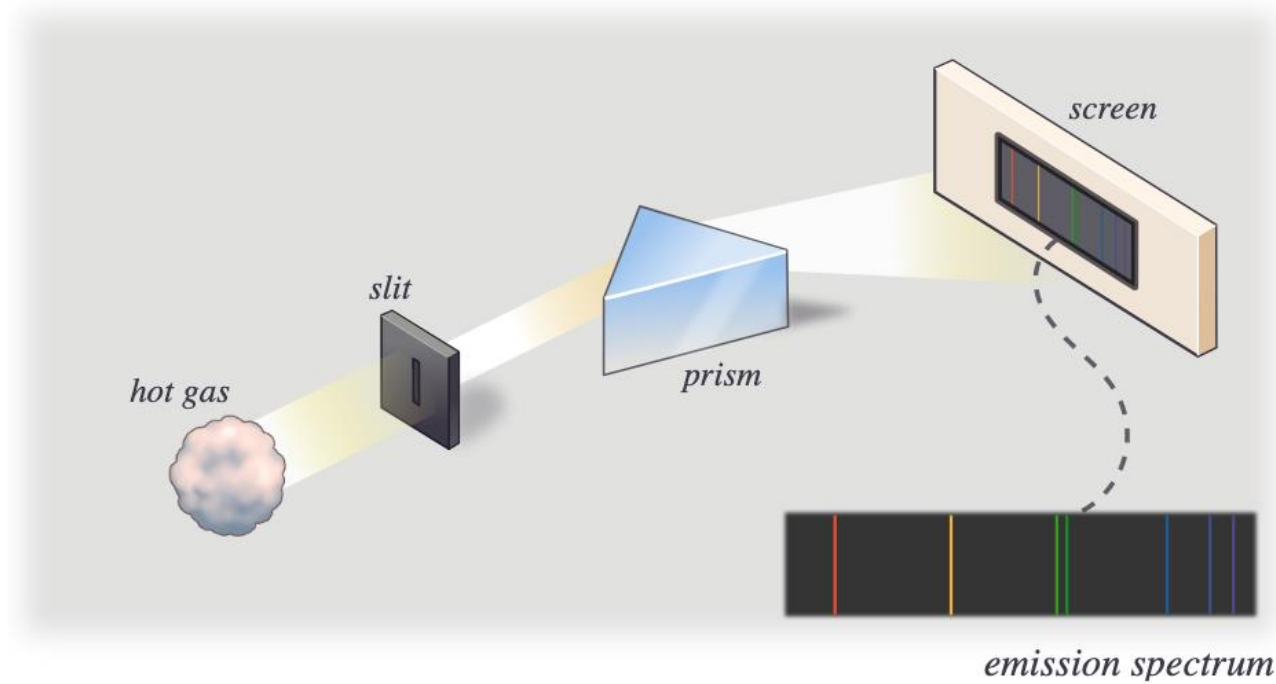
The classical theory predicts that electromagnetic radiation is continuous

Hydrogen Atomic Spectrum



An absorption spectrum results when light from a continuous source passes through a cooler gas, consisting of a series of dark lines characteristic of the composition of the gas.

Hydrogen Atomic Spectrum



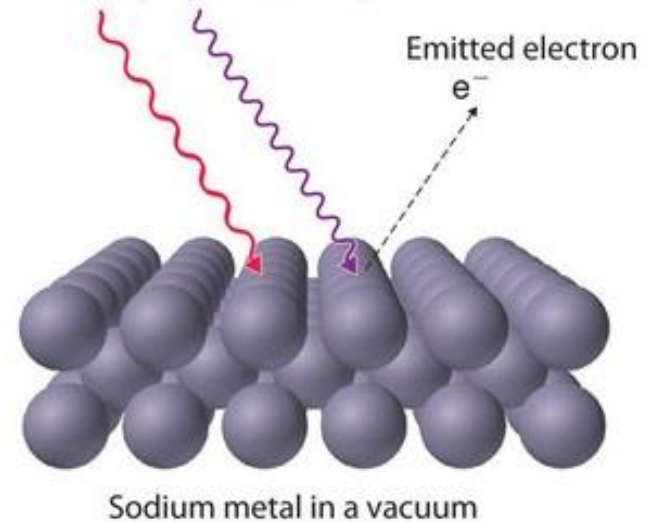
- When a high-energy discharge is passed through a sample of H_2 gas, the H_2 molecules absorb energy, causing some of the H-H bonds to break.
- The resulting hydrogen atoms are *excited*; they contain excess energy, which they release by emitting light of various wavelengths to produce what is called the *emission spectrum* of the hydrogen atom., called a **line spectrum**.

The Photoelectric Effect

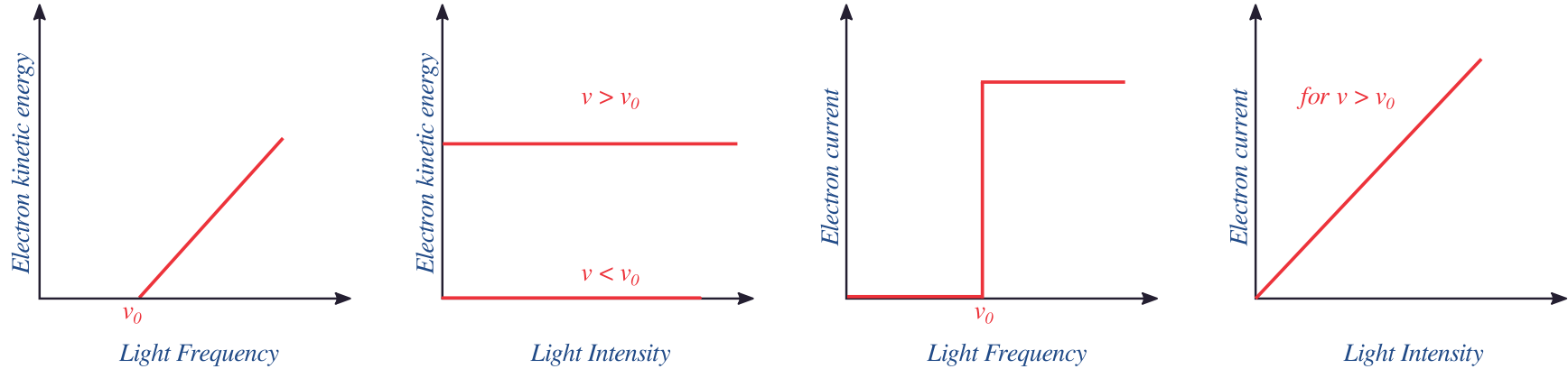
A beam of light hitting a metal surface can cause electrons to be ejected from the surface.

Classical Paradigm: the energy of the ejected electrons should be proportional to the intensity (I) of the light and **independent of the frequency (ν) of the light.**

Experiment: the energy of the ejected electrons is independent of the intensity (I) and **depends directly on the frequency (ν) of the light.**



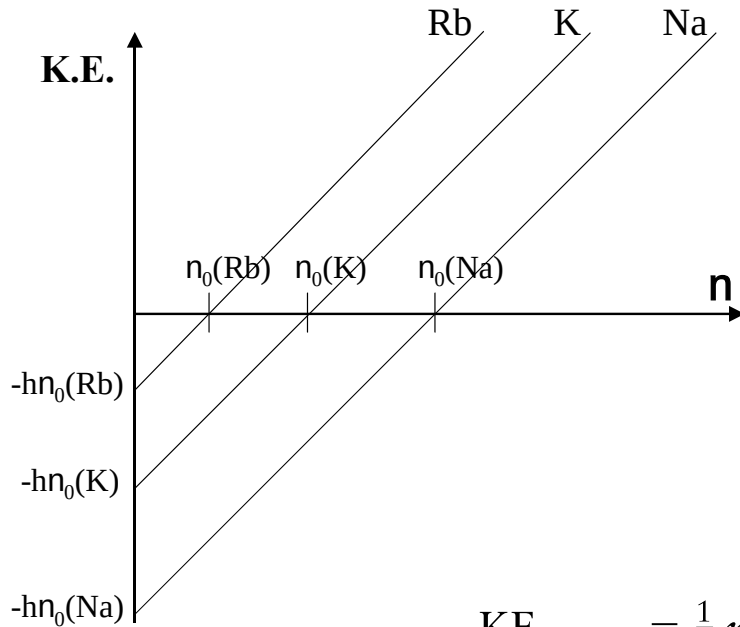
The Photoelectric Effect



The following observations characterize the photoelectric effect:

1. No electrons are emitted by a given metal below a specific threshold frequency ν_0 .
2. For light with frequency lower than the ν_0 , no electrons are emitted regardless of the intensity of the light. For light with frequency greater than the ν_0 , the kinetic energy of the emitted electrons increases linearly with the frequency of the light and remains independent of intensity.
3. The number of electrons emitted per second (i.e. the electric current) is independent of light frequency above the ν_0 and zero below.
4. For light with frequency greater than the ν_0 , the number of electrons emitted increases with the intensity of the light.

Quantization of Light

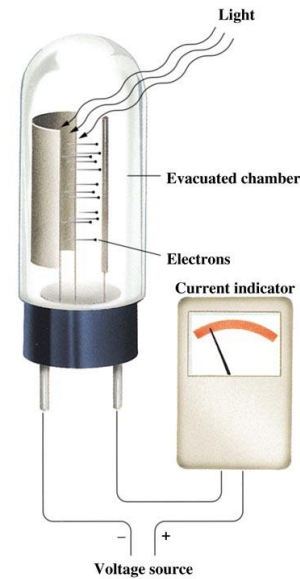


The slope of KE vs ν is h

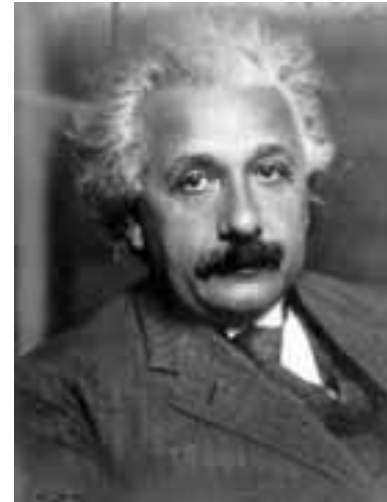
Workfunction, ϕ

$$KE_{\text{electron}} = \frac{1}{2}mv^2 = h\nu - h\nu_0$$

$\nearrow$ Mass of electron $\uparrow$ Velocity of electron $\uparrow$ Energy of incident photon $\nwarrow$ Energy required to remove electron from metal's surface



© 2003 Thomson-Brooks/Cole



Quantization of Light

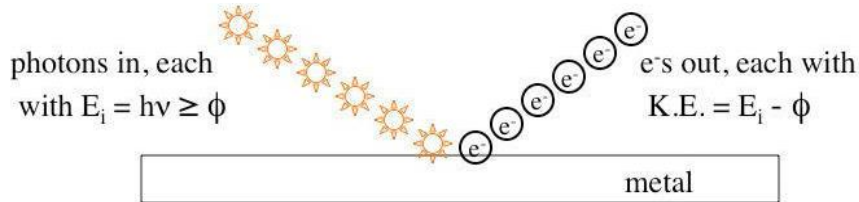
- These observations can be explained by assuming that electromagnetic radiation is quantized (consists of photons) and that the threshold frequency represents the minimum energy required to remove the electron from the metal's surface.
- Minimum energy required to remove an electron, $E_0 = h\nu_0$



Light as a wave phenomenon

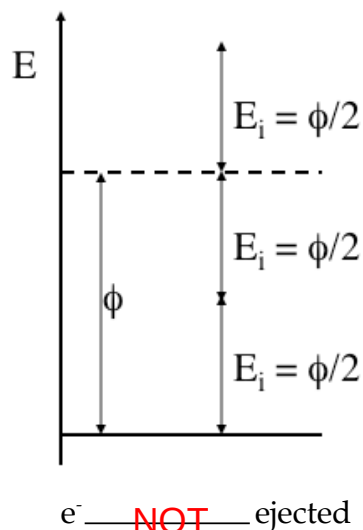
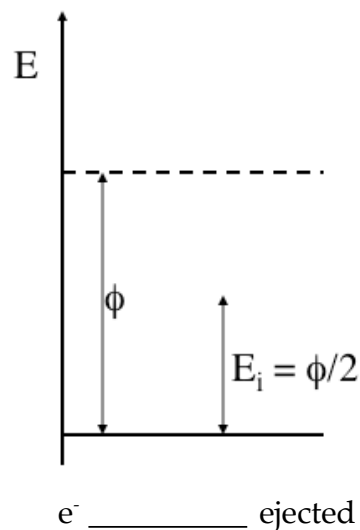
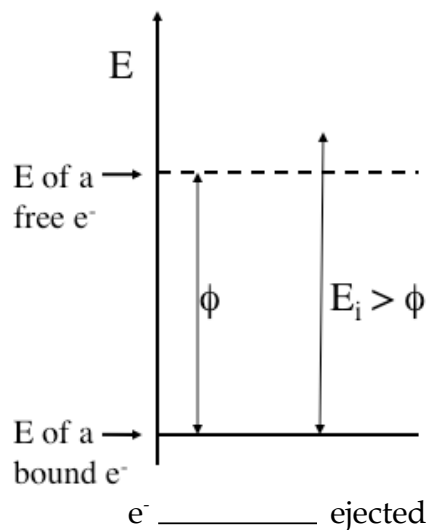
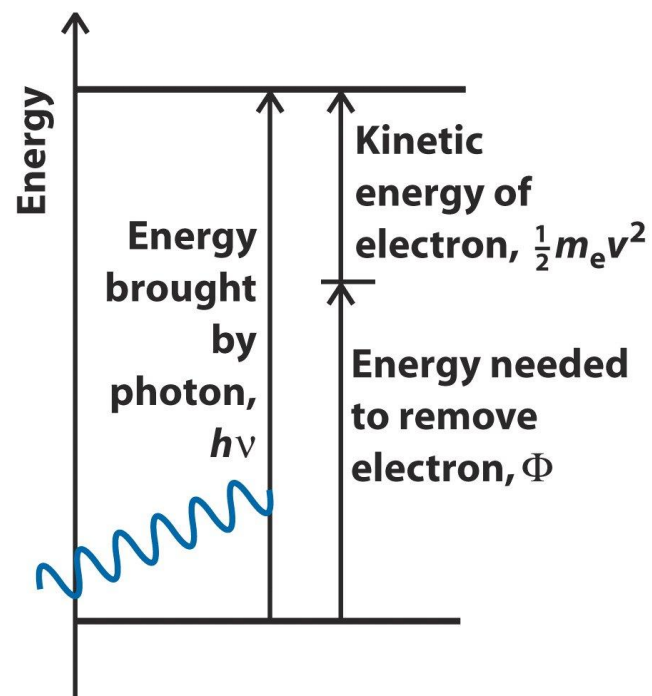


Light as a stream of photons



$$E = \frac{nhc}{\lambda} \quad (n = 0, 1, 2, 3, \dots)$$

Quantization of Light



Waves behaving as Particles

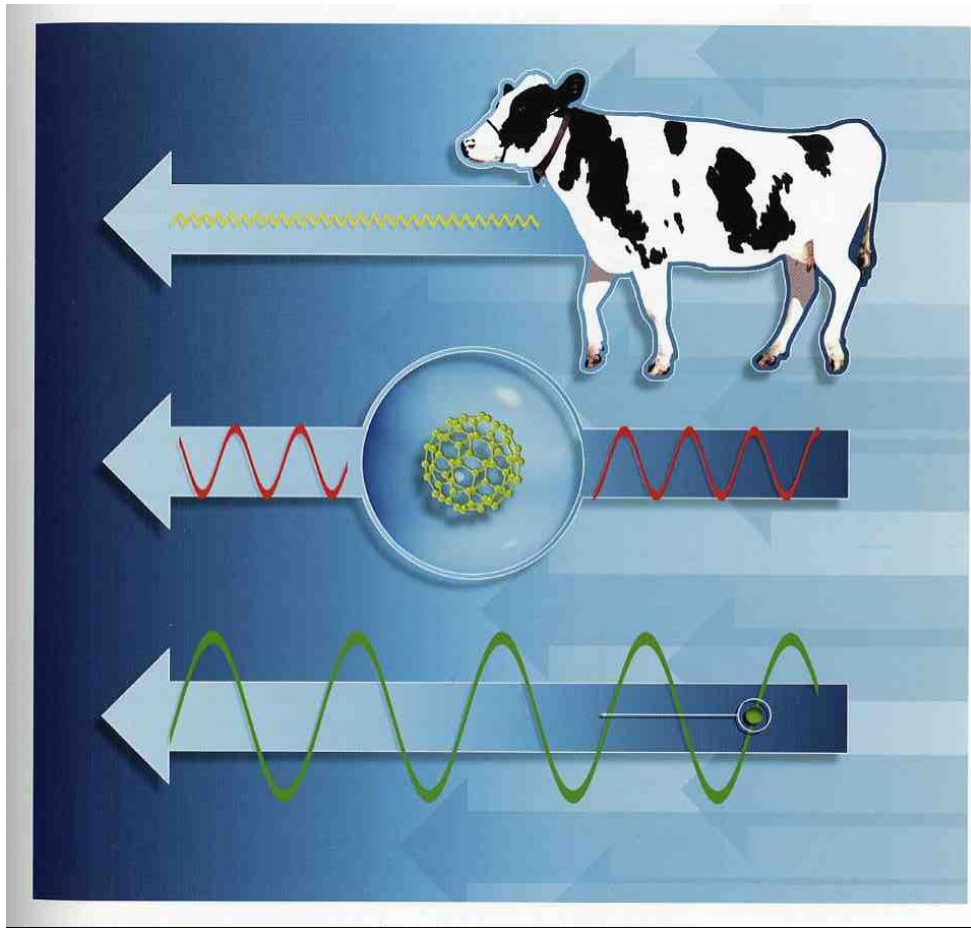
Wave Nature of Particles



- Louis de Broglie **postulated** that as *light has wave-like and particle-like properties*, *matter must also be both particle-like and wave-like*.
- A particle, of mass **m**, travelling at velocity **v**, has linear momentum **p = mv**.
- By analogy with photons, the associated wavelength of the particle (**λ**) is given by:

de Broglie wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$ de Broglie's equation

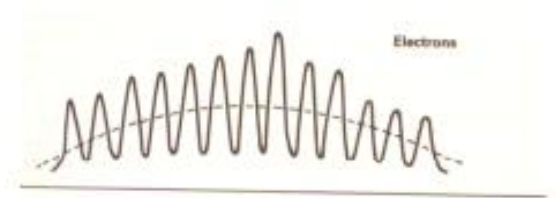
The fact that particles can behave as waves but also as particles, depending on which experiment you perform on them, is known as the *particle-wave duality*.



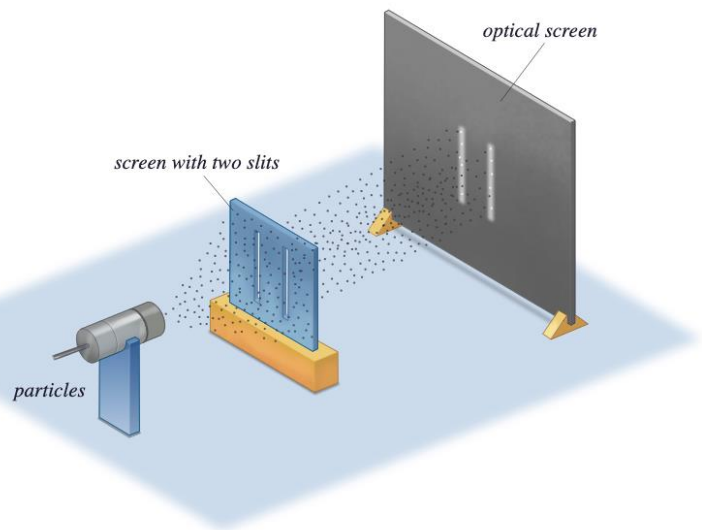
The wave properties of matter are only apparent for very small masses of matter.



- Bullets are far more massive than the electrons. One can observe them as long as one likes but it would not make any difference to them.
- The interference wiggles in the case of bullets are so crowded that it is physically impossible to resolve them, one sees an average behaviour



Wave Nature of Particles: Validation

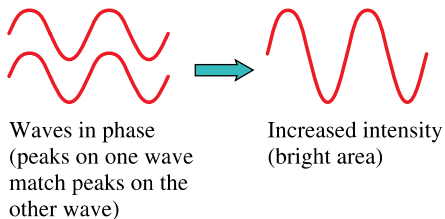


According to classical physics, electrons should behave like particles - they travel in straight lines and do not curve in flight unless acted on by an external agent.

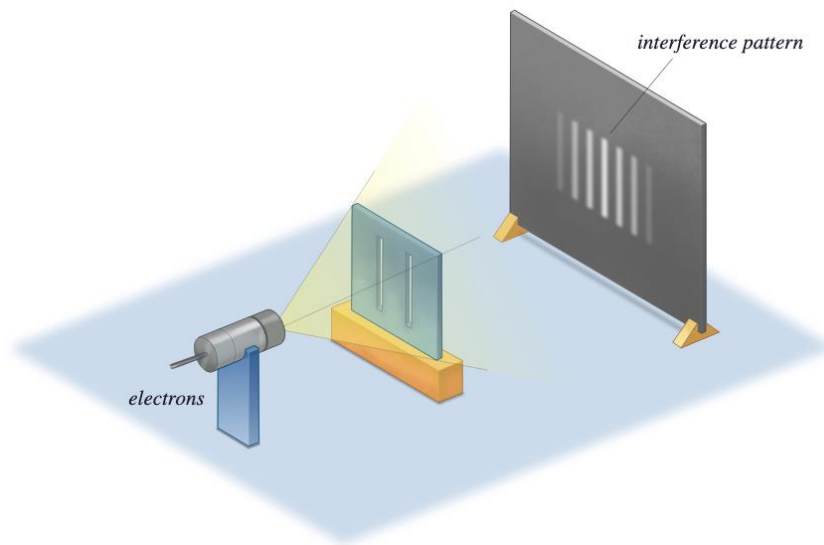
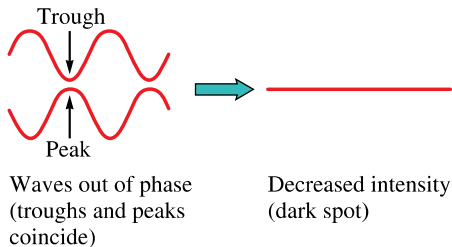
In this model, if we fire a beam of electrons through a double slit onto a detector, we should get two bands of "hits".

Can be explained only in terms of waves.

(b) Constructive interference



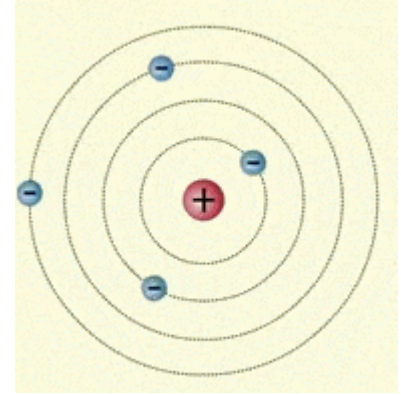
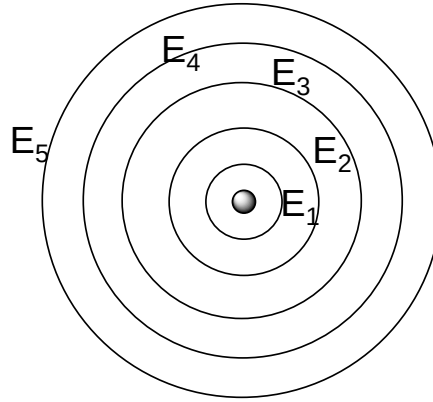
(c) Destructive interference



Wave-Particle Duality

- Now we have come full circle.
- Electromagnetic radiation, which at the turn of the twentieth century was thought to be a pure waveform, was found to exhibit particulate properties.
- Conversely, electrons, which were thought to be particles, were found to have a wavelength associated with them.
- Energy is really a form of matter, and all matter shows the same types of properties.
- *All matter exhibits both particulate and wave properties.*

Putting it All together: The Bohr Atom

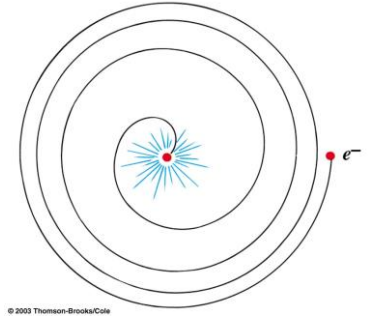


- Bohr proposed a model that included the idea that the *electron in a hydrogen atom moves around the nucleus **ONLY** in certain allowed circular orbits.*
- Bohr assumed that the hydrogen electron could ***exist only in stationary, non-radiating orbits.***

Putting it All together: The Bohr Atom

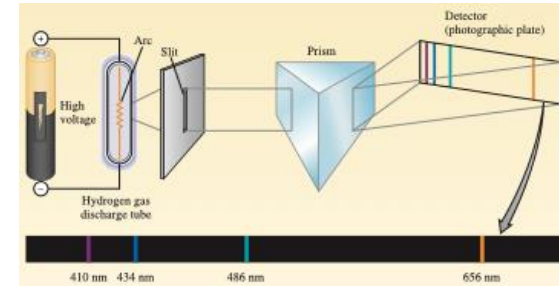
Solving Paradox I: The unstable orbiting electron of Rutherford's atom.

- Make the orbiting electron stable by assuming that the *electron's orbit and energy is quantized to certain values and for these values the orbiting electron does not radiate.*
- The electron is stable in these orbits.
- Thus, the *orbits and energies of electrons are quantized.*

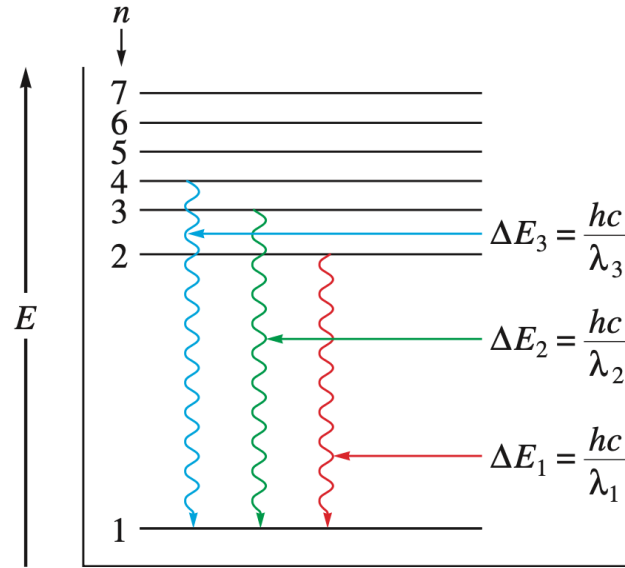
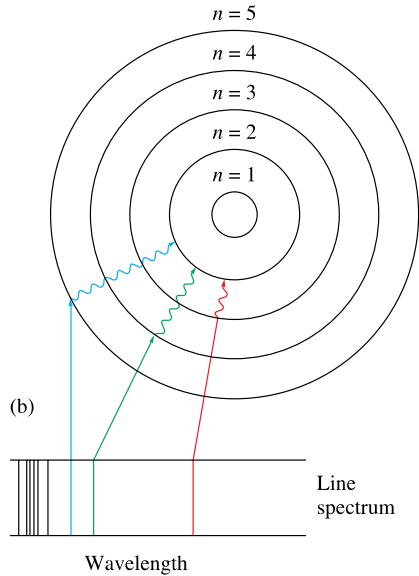


Solving Paradox II: The line spectra of emitting or absorbing atoms.

- Since only certain energies are allowed for orbiting electrons, *only jumps between orbits can be observed.*
- These jumps correspond to *discrete frequencies and wavelengths.*
- Thus, line spectra as expected because of the quantized energies of the orbits.



The Bohr Atom



- Light is **emitted** when an electron jumps from a higher orbit to a lower orbit and **absorbed** when it jumps from a lower to higher orbit.
- The energy and frequency of light emitted or absorbed is given by the difference between the two orbit energies, e.g.,
- $E(\text{photon}) = E_2 - E_1$ (Energy difference) $= h\nu$

$$\Delta E = h\nu = \frac{hc}{\lambda}$$

Energy of an electron

$$E(n) = -\frac{1}{n^2} \cdot 13.6 \text{ eV}$$

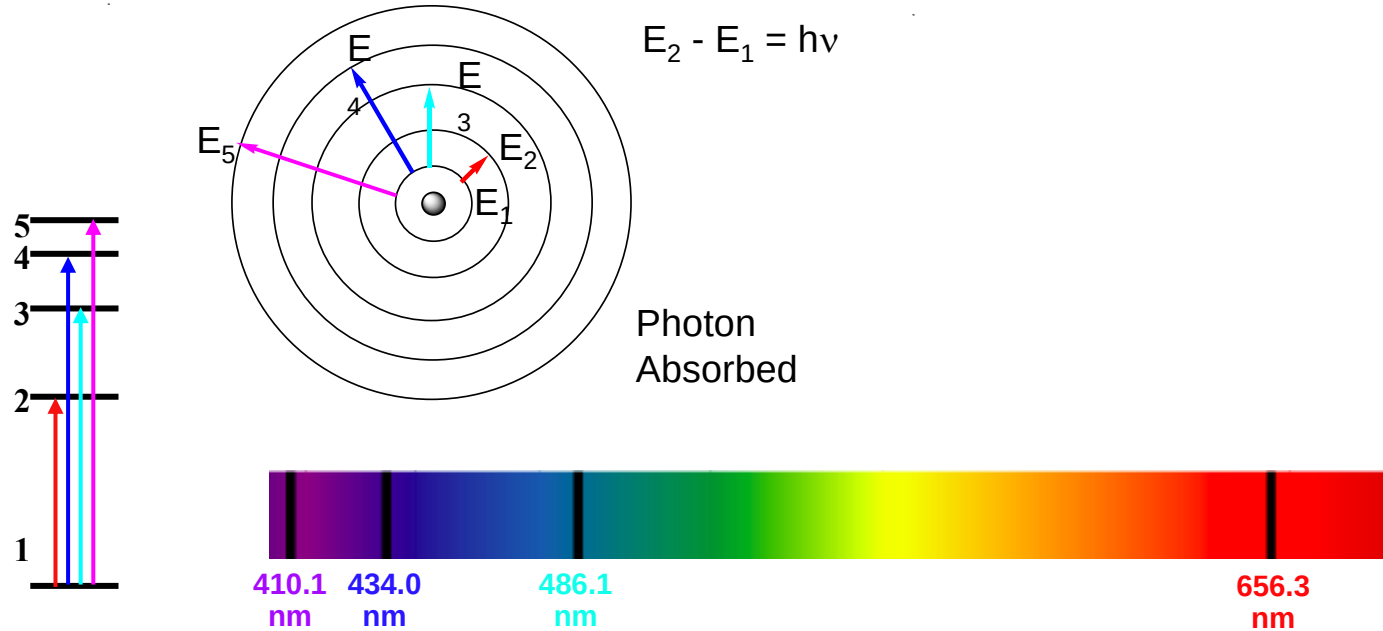
$$E_n = -\frac{Z^2 R_E}{n^2}.$$

Transition Energy

$$\Delta E = E_f - E_i = -R_E \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

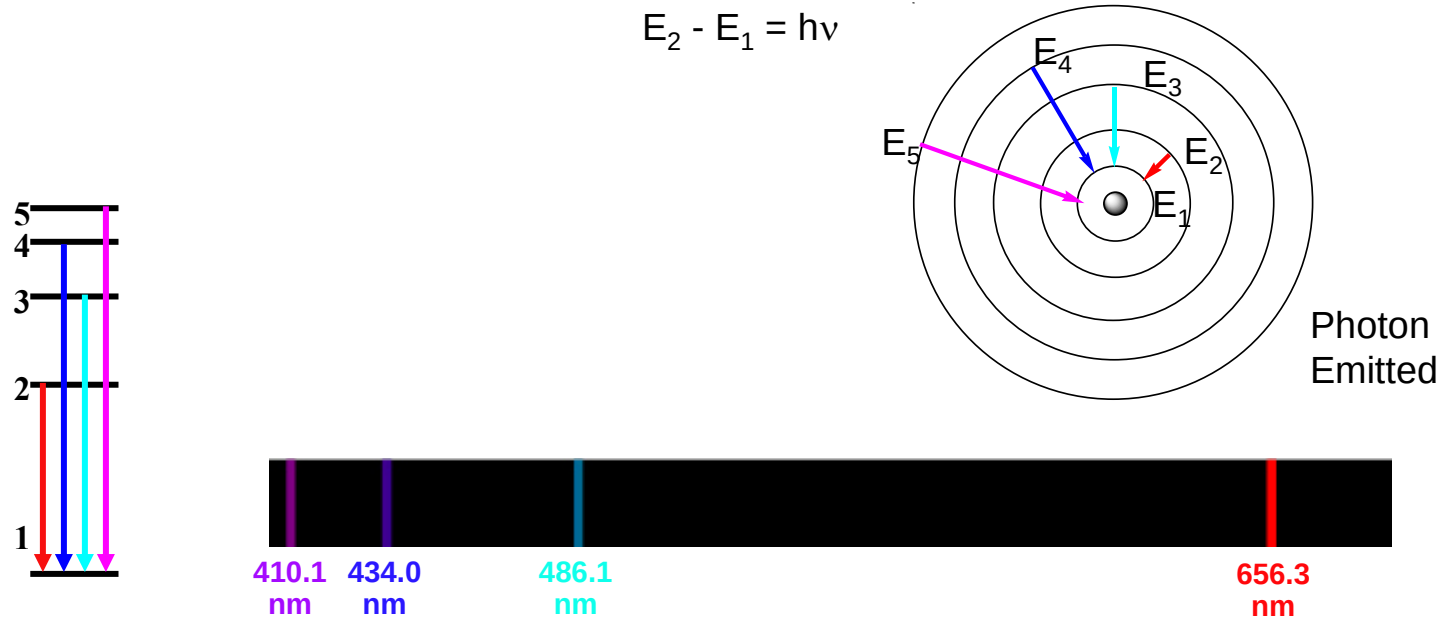
R_E : Rydberg constant and its value is $2.18 \times 10^{-18} \text{ J}$

The Bohr Atom



Bohr atom: Light absorption occurs when an electron **absorbs a photon** and makes a transition for a lower energy orbital to a higher energy orbital. **Absorption spectra appear as sharp lines.**

The Bohr Atom



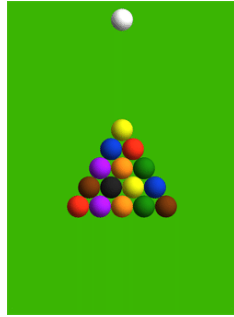
Bohr atom: Light emission occurs when an electron makes a transition from a higher energy orbital to a lower energy orbital and a **photon is emitted**. Emission spectra appear as sharp lines.

The Bohr Atom

Limitations of Bohr Model

- The model only works for hydrogen (and other one electron ions) – ignores e-e repulsion.
- When Bohr's model was applied to atoms other than hydrogen, it did not work
- Does not explain fine structure of spectral lines.
- **Note:** *The Bohr model (assuming circular electron orbits) is fundamentally incorrect.*

Heisenberg's Uncertainty Principle



- Bohr orbit: The electron in the orbital is moving around the nucleus in a circular orbit.
- We can predict the motion of the electron, similar to the motions of particles in the macroscopic world.
- For example, when two billiard balls with known velocities collide, we can predict their motions after the collision.
- Werner Heisenberg, discovered a very important principle in 1927—the **Heisenberg uncertainty principle**.

Heisenberg's Uncertainty Principle

- The Heisenberg Uncertainty Principle is a fundamental theory in quantum mechanics that defines why a scientist **cannot measure multiple quantum variables simultaneously**.
- *There is a fundamental limitation to just how precisely we can know both the position and the momentum of a particle at a given time.*

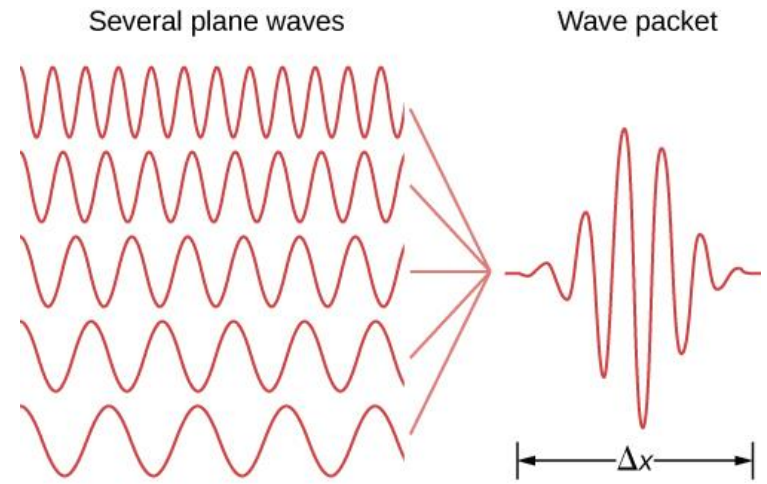
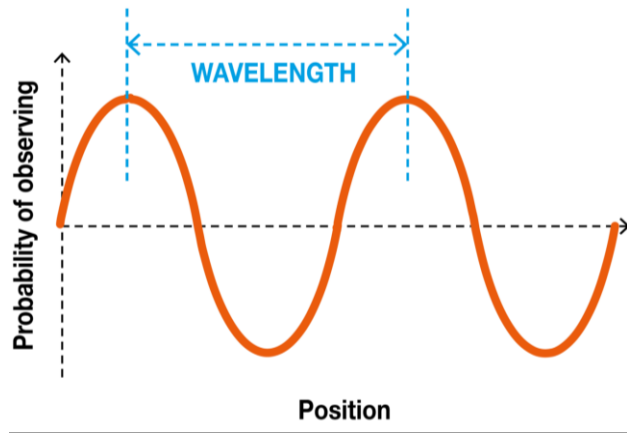
$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

$$\hbar = h/2\pi$$

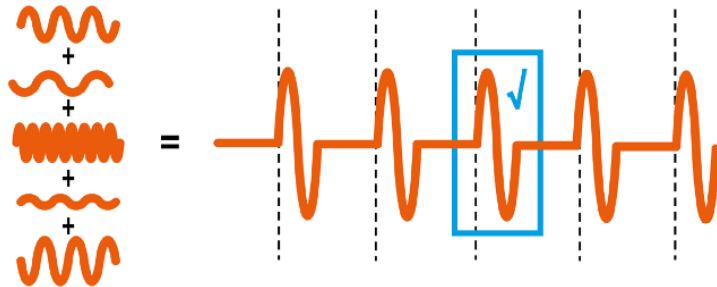
where Δx is the uncertainty in a particle's position, Δp is the uncertainty in a particle's momentum.

Note: There is no restriction on the precision in simultaneously knowing/measuring the position along a given direction (**x**) and the momentum along another, perpendicular direction (**z**):

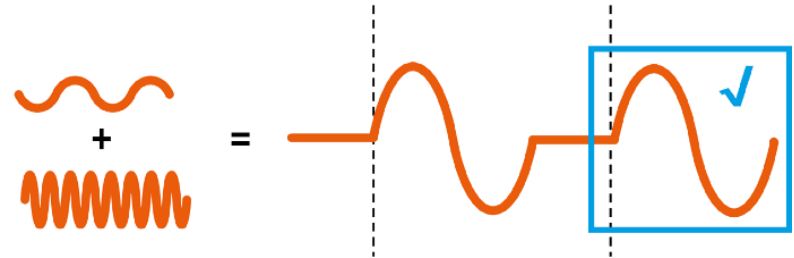
$$\Delta p_z \cdot \Delta x = 0$$



**DECREASING POSITION UNCERTAINTY
=
INCREASING MOMENTUM UNCERTAINTY**



**DECREASING MOMENTUM UNCERTAINTY
=
INCREASING POSITION UNCERTAINTY**



- This limitation is so small for large particles such as baseballs or billiard balls that it is unnoticed.
- However, for a small particle such as the electron, the limitation becomes quite important.
- Applied to the electron, the uncertainty principle implies that we cannot know the exact path of the electron as it moves around the nucleus.
- *It is therefore not appropriate to assume that the electron is moving around the nucleus in a well-defined orbit as in the Bohr model.*

Quantum Mechanics

Schrödinger: If electrons are waves, their position and motion in space must obey a wave equation.

Solutions of wave equations yield **wavefunctions**, Ψ , which contain the information required to describe ALL of the properties of the wave.



The Schrödinger Equation (1926)

- The central equation in Quantum Mechanics.
- Observable = total energy of system.

Schrödinger Equation

$$\hat{\mathbf{H}}\psi = E\psi$$

$\hat{\mathbf{H}}$ Hamiltonian Operator

E Total Energy

where $\hat{\mathbf{H}} = \hat{\mathbf{T}} + \hat{\mathbf{V}}$ and $E = T + V$.

- SE can be set up for any physical system.
- The form of $\hat{\mathbf{H}}$ depends on the system.
- Solve SE $\Rightarrow \psi$ and corresponding E .

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right] \psi(\vec{r}) = E\psi(\vec{r})$$

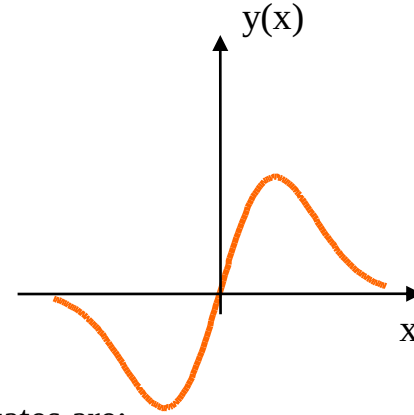
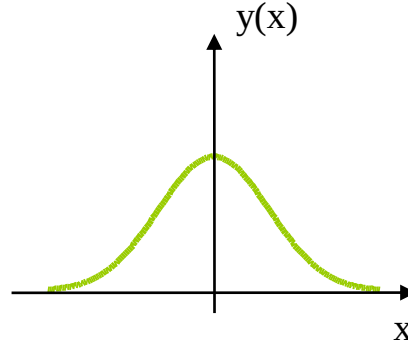
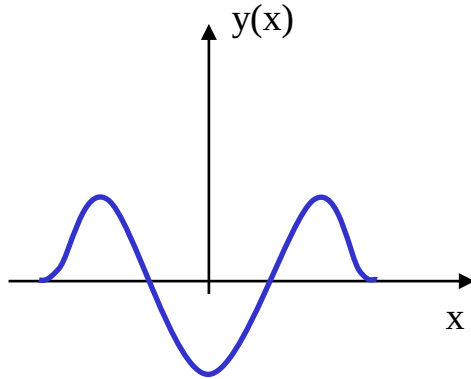
Wavefunctions

- In QM, a “particle” is distributed in space like a wave.
- We cannot define a position for the particle.
- Instead, we define a probability of finding the particle at any point in space.
- **Wavefunction** (ψ) – a wave representing the spatial distribution of a “particle”.
- $|\psi(x)|$ – the magnitude of ψ at point x .
- Because ψ may be imaginary or complex $\Rightarrow \psi^2$ would be negative or complex.
- For the general case, where ψ is complex ($\psi = a + ib$) then:
 $|\psi|^2 = \psi^* \psi$ where ψ^* is the complex conjugate of ψ .
probability density ($\psi^* = a - ib$)

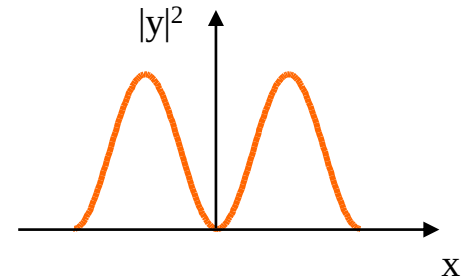
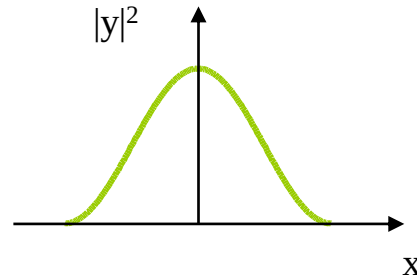
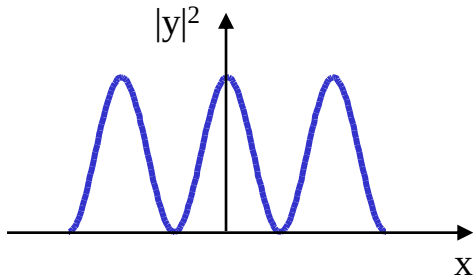
$$i = \sqrt{-1}$$

Particle Wavefunctions: Examples

Examples of $\psi(x)$

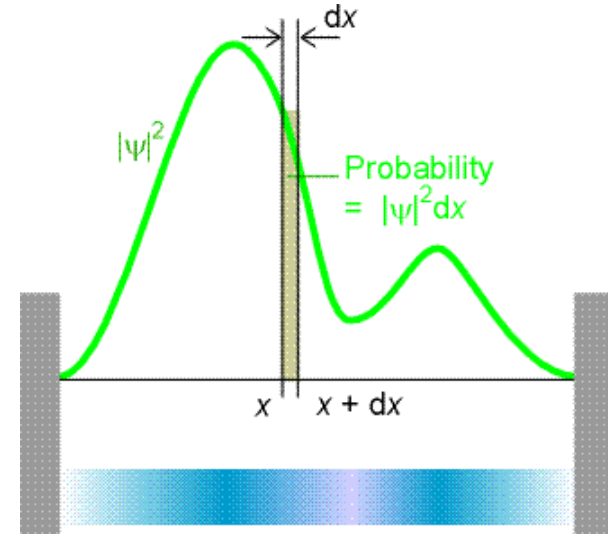


The corresponding probability distributions $|\psi(x)|^2$ of these states are:



Key point: Particle cannot be associated with a specific location x .

Wavefunctions



Example: 1-D System

If the wavefunction at point x is $\Psi(x)$, the probability of finding the particle in the infinitesimally small region (dx) between x and $x+dx$ is:

$$P(x) \propto |\Psi(x)|^2 dx$$

Normalization of the Wavefunction

$$P(\mathbf{r}) \propto |\psi(\mathbf{r})|^2 d\tau$$

- If ψ is such that the sum of $|\psi(\mathbf{r})|^2$ at all points in space = 1, then:

$$P(x) = |\psi(x)|^2 dx \quad \text{1-D}$$

$$P(\mathbf{r}) = |\psi(\mathbf{r})|^2 d\tau \quad \text{3-D}$$

- As summing over an infinite number of infinitesimal steps = **integration**, this means:

$$P_{\text{total}}(1D) = \int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$$

$$P_{\text{total}}(3D) = \int_{-\infty}^{\infty} |\psi(\mathbf{r})|^2 d\tau = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |\psi(x, y, z)|^2 dx dy dz = 1$$

- i.e. the probability that the particle is somewhere in space = 1.
- In this case, ψ is said to be a **normalized wavefunction**.

How to Normalize the Wavefunction

- If ψ is not normalized, then:

$$\int |\psi(\mathbf{r})|^2 d\tau = A \neq 1$$

- A corresponding normalized wavefunction (ψ_{Norm}) can be defined:

$$\psi_{\text{Norm}}(\mathbf{r}) = \frac{1}{\sqrt{A}} \psi(\mathbf{r})$$

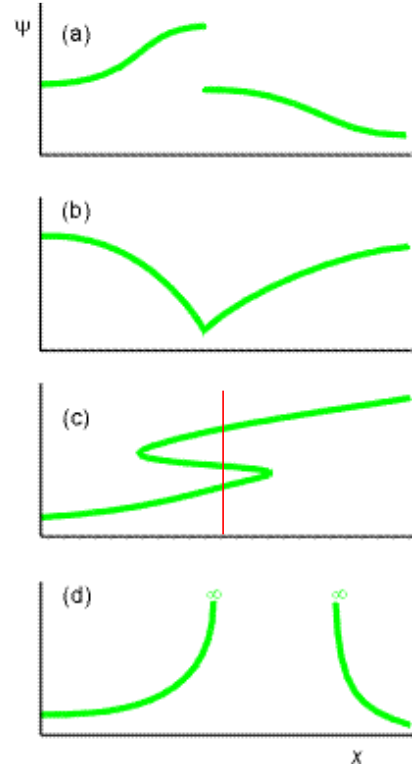
such that:

$$\int |\psi_{\text{Norm}}(\mathbf{r})|^2 d\tau = 1$$

- The factor ($1/\sqrt{A}$) is known as the **normalization constant** (sometimes represented by **N**).

Properties of Acceptable Wavefunctions

- Ψ must be finite everywhere
- Ψ must be single-valued
- Ψ and $d\Psi/dx$ must be continuous for finite well-behaved potentials $V(x)$
- Ψ must be normalizable (generally $\Psi \rightarrow 0$ as $x \rightarrow \pm\infty$)



Wavefunctions

The Importance of ψ

- ψ completely defines the system (e.g. electron in an atom or molecule).
- If ψ is known, we can determine any observable property (e.g. energy, vibrational frequencies, ...) of the system.
- QM provides the tools to determine ψ computationally, to interpret ψ and to use ψ to determine properties of the system.

Hamiltonian Operator

- Wavefunction (ψ) contains all the information we need to know about any particular system.
- How do we determine ψ and use it to deduce properties of the system?

Operators and Observables

- If ψ is the wavefunction representing a system, we can write:

$$\hat{Q}\psi = Q\psi$$

where Q – “observable” property of system (e.g. energy, momentum, dipole moment ...)

$\hat{Q}$ – operator corresponding to observable Q .

$$d/dx (e^{ax}) = a e^{ax}$$

$$d^2/dx^2 (\sin ax) = -a^2 \sin ax$$

$$\hat{x} = x$$

Operator for position in the x-direction is just multiplication by x

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$$

Operator for linear momentum in the x-direction:

Constructing Kinetic and Potential Energy QM Operators

1. Write down classical expression in terms of position and momentum.
2. Introduce QM operators for position and momentum.

Examples

1. Kinetic Energy Operator in 1-D $\hat{T}_x$

$$\text{CM} \quad T_x = \frac{p_x^2}{2m} \quad \Rightarrow \quad \text{QM} \quad \hat{T}_x = \frac{\hat{\mathbf{p}}_x^2}{2m} = -\frac{\hbar^2}{2m} \left(\frac{d^2}{dx^2} \right)$$

2. KE Operator in 3-D $\hat{T}$

$$\text{CM} \quad T = \frac{p^2}{2m} = \frac{p_x^2 + p_y^2 + p_z^2}{2m} \quad \text{QM} \quad \hat{T} = \frac{\hat{\mathbf{p}}^2}{2m} = -\frac{\hbar^2}{2m} \left(\underbrace{\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}}_{\substack{\text{partial derivatives} \\ \text{operate on } \psi(x,y,z)}} \right) = -\frac{\hbar^2}{2m} \nabla^2$$

“del-squared”
↓

3. Potential Energy Operator $\hat{V}$ (a function of position)

∴ PE operator corresponds to multiplication by $V(x)$, $V(x,y,z)$ etc.

Examples

1. Particle Moving in 1-D $\Psi(x)$

$$\hat{\mathbf{H}}\psi = \hat{\mathbf{T}}\psi + \hat{\mathbf{V}}\psi = E\psi \longrightarrow -\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}\right) + V(x)\psi = E\psi$$

- The form of $V(x)$ depends on the physical situation:
 - Free particle $V(x) = 0$ for all x .
 - Harmonic oscillator $V(x) = \frac{1}{2}kx^2$

2. Particle Moving in 3-D $\Psi(x,y,z)$

- SE $\Rightarrow -\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2}\right) + V(x, y, z)\psi = E\psi$

or
$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(x, y, z)\psi = E\psi$$

Note: The SE is a second order differential equation

Time-Dependent Schrodinger Wave Equation

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \Psi(x, t) + V(x) \Psi(x, t)$$

PHYSICS NOTATION

↑ Total E term ↑ K.E. term ↑ P.E. term

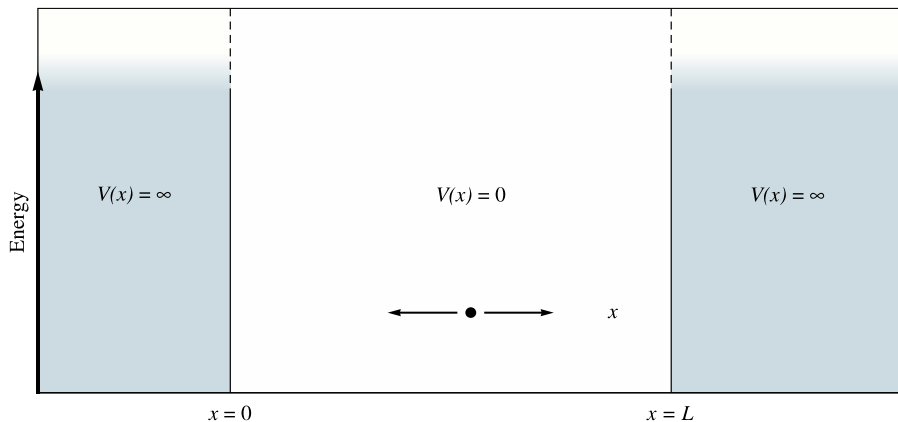
$$\Psi(x, t) = e^{-iEt/\hbar} \psi(x)$$

Time-Independent Schrodinger Wave Equation

$$E\psi(x) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x) + V(x)\psi(x)$$

Particle in a Box model

- Consider a particle with mass m that is free to move back and forth along one dimension between the values $x = 0$ and $x = L$.
- The potential energy $V(x)$ of the particle is zero at all points along its path, except at the endpoints $x = 0$ and $x = L$, where $V(x)$ is infinitely large.
- Schrödinger equation contains the energy operator $\hat{H}$.
- The potential energy is zero inside the box, the only energy possible is the kinetic energy of the particle as it moves back and forth along the x axis.



$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \quad \rightarrow \quad \frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}\psi$$

Particle in a Box model

- Each solution must be a function whose second derivative has the same form as the original function.
- One function that behaves this way is the sine function.

$$\begin{aligned}\frac{d^2}{dx^2}(A \sin kx) &= A \frac{d}{dx} \left(\frac{d \sin kx}{dx} \right) = A \frac{d}{dx} (k \cos kx) \\ &= Ak \left(\frac{d \cos kx}{dx} \right) = Ak(-k \sin kx) \\ &= -Ak^2 \sin kx = -k^2 A \sin kx\end{aligned}$$

Restrictions on ψ

- In principle **Schrödinger Eqn.** has an infinite number of solutions.
- So far we have general solutions:
 - any value of $\{A, k\} \rightarrow$ any value of $\{\psi, E\}$.
- **BUT** – due to the **Born interpretation of ψ** , only certain values of ψ are physically acceptable:
 - outside box ($x < 0, x > L$) $V = \infty \Rightarrow$ impossible for particle to be outside the box
$$\therefore |\psi|^2 = 0 \Rightarrow \psi = 0 \text{ outside box.}$$
 - ψ must be a continuous function
$$\Rightarrow \text{Boundary Conditions} \quad \begin{array}{l} \psi = 0 \text{ at } x = 0 \\ \psi = 0 \text{ at } x = L. \end{array}$$

Effect of Boundary Conditions

1. $x = 0$

$$\psi = A \sin kx$$

$$\psi = 0$$

$\Rightarrow$

$$\psi = A \sin kx$$

for all x

2. $x = L$

$$\psi = A \sin kL = 0$$

$A=0$? $\times$
(or $\psi = 0$ for all x)

$\sin kL = 0$?

✓

$$\sin kL = 0$$

$\Rightarrow$

$$kL = n\pi$$

$$n = 1, 2, 3, \dots$$

($n \neq 0$, or $\psi = 0$ for all x)

Allowed Wavefunctions and Energies

- k is restricted to a discrete set of values: $k = n\pi/L$
- Allowed wavefunctions: $\psi_n = A \sin(n\pi x/L)$

Total probability of finding
the particle in the box $= \int_0^L \psi^2(x) dx = 1$

$$\int_0^L \psi^2(x) dx = \int_0^L A^2 \sin^2\left(\frac{n\pi}{L}x\right) dx = 1$$

$$\int_0^L \sin^2\left(\frac{n\pi}{L}x\right) dx = \frac{1}{A^2}$$

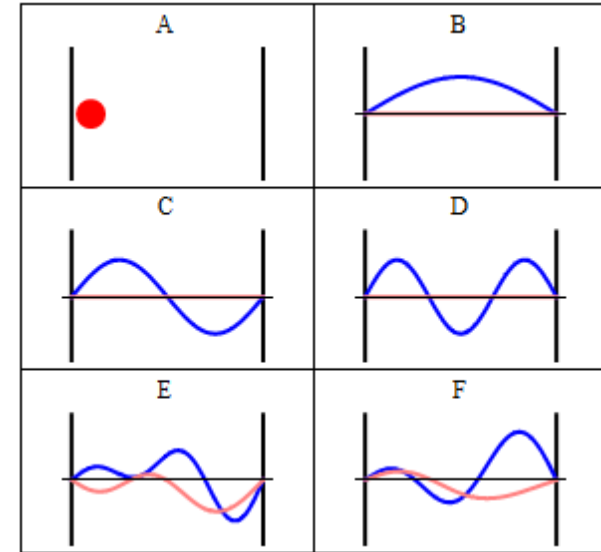
$$\frac{L}{2} = \frac{1}{A^2} \quad \text{and} \quad A = \sqrt{\frac{2}{L}}$$

- Normalization: $A = \sqrt{2/L} \Rightarrow \psi_n = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$

Allowed Wavefunctions and Energies

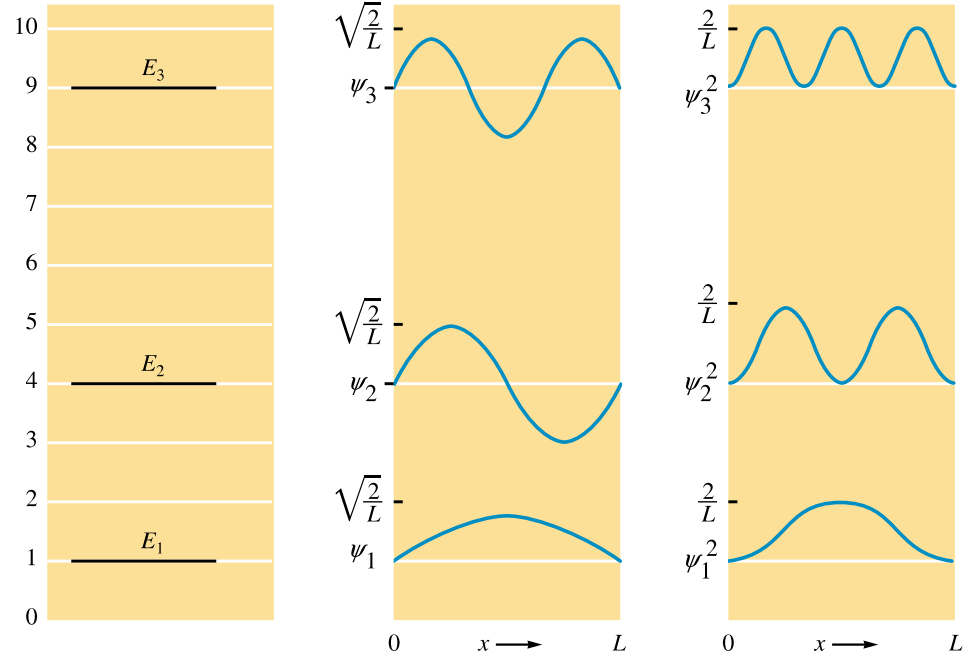
- Normalization: $\Rightarrow \psi_n = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$

- Allowed energies:
$$E_n = \frac{n^2 h^2}{8mL^2}$$



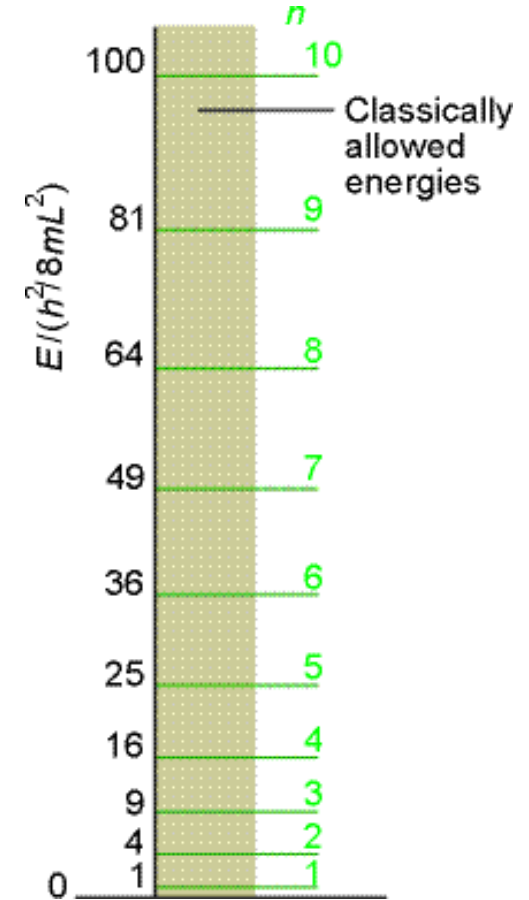
n	Function	Energy
1	$\psi_1 = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi}{L}x\right)$	$E_1 = \frac{h^2}{8mL^2}$
2	$\psi_2 = \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi}{L}x\right)$	$E_2 = \frac{4h^2}{8mL^2} = \frac{h^2}{2mL^2}$
3	$\psi_3 = \sqrt{\frac{2}{L}} \sin\left(\frac{3\pi}{L}x\right)$	$E_3 = \frac{9h^2}{8mL^2}$
4	$\psi_4 = \sqrt{\frac{2}{L}} \sin\left(\frac{4\pi}{L}x\right)$	$E_4 = \frac{16h^2}{8mL^2} = \frac{2h^2}{mL^2}$
$\vdots$	$\vdots$	$\vdots$

The application of the boundary conditions has led to a series of *quantized* energy levels. That is, only certain energies are allowed for the particle bound in the box.



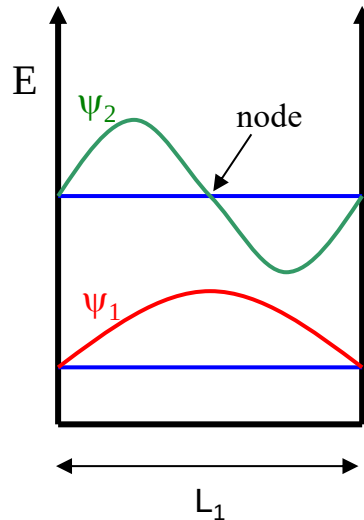
Quantum Numbers

- There is a discrete energy state (E_n), corresponding to a discrete wavefunction (ψ_n), for each integer value of n .
- Quantization – occurs due to boundary conditions and requirement for ψ to be physically reasonable (Born interpretation).
- n is a **Quantum Number** – labels each allowed state (ψ_n) of the system and determines its energy (E_n).
- Knowing n , we can calculate ψ_n and E_n .

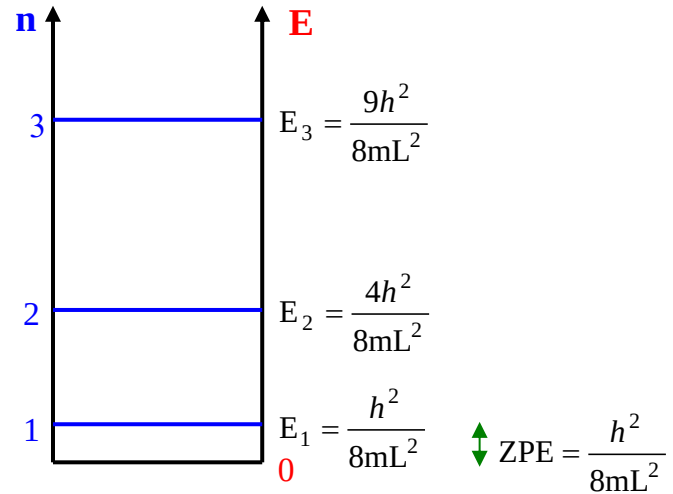


Properties of the Wavefunction

- **Nodes** in the wavefunction – points at which $\psi_n = 0$ (excluding the ends which are constrained to be zero).
- Number of nodes = $(n-1)$ $\psi_1 \rightarrow 0; \psi_2 \rightarrow 1; \psi_3 \rightarrow 2 \dots$



- $E_n \propto n^2 \Rightarrow$ energy levels get further apart as $n \uparrow$



- Zero-Point Energy (ZPE) – lowest energy of particle in box:

$$ZPE = E_{\min} = E_1 = \frac{h^2}{8mL^2}$$

- CM $E_{\min} = 0$
- QM $E = 0$ corresponds to $\psi = 0$ everywhere (forbidden).

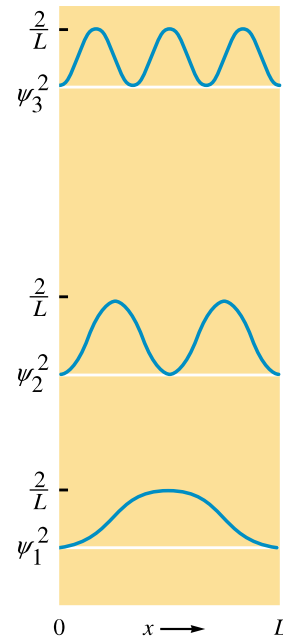
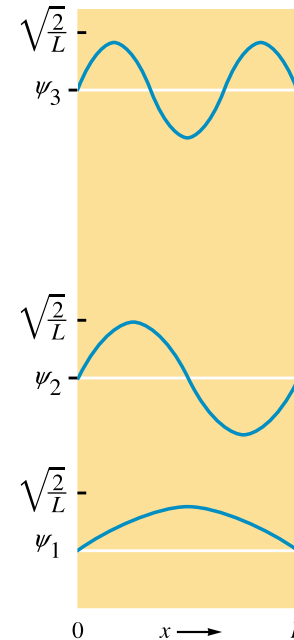
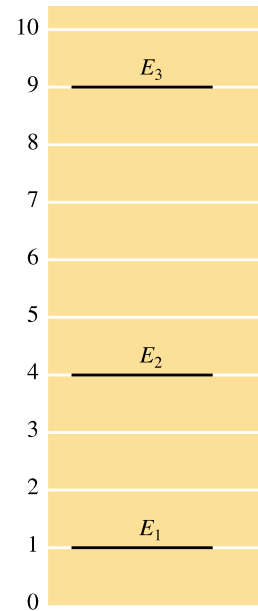
Density Distribution of the Particle in the 1-D Box

- The probability of finding the particle between x and $x+dx$ (in the state represented by ψ_n) is:

$$P_n(x) = |\psi_n(x)|^2 dx = (\psi_n(x))^2 dx \quad (\psi_n \text{ is real})$$

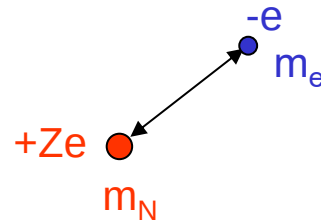
$$\Rightarrow P_n(x) = \frac{2}{L} \sin^2\left(\frac{n\pi x}{L}\right) dx$$

- Note:** probability is not uniform
 - $\psi_n^2 = 0$ at **walls** ($x = 0, L$) for all ψ_n .
 - $\psi_n^2 = 0$ at **nodes** (where $\psi_n = 0$).



Hydrogen Atom

- The series of atoms/ions $\text{H}, \text{He}^+, \text{Li}^{2+}, \text{Be}^{3+} \dots$
all have: 1 electron (charge $-e$)
 nucleus (charge $+Ze$)
- To determine the electronic wavefunction ($\psi = \psi_e$) and allowed electronic energy levels (E), we must set up and solve the Schrödinger Equation for a single electron in an atom.



- Consider 3-D motion of the electron ($-e$) relative to the nucleus ($+Ze$):

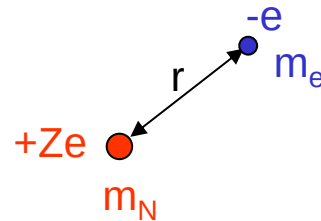
Hamiltonian Operator $\hat{\mathbf{H}} = \hat{\mathbf{T}} + \hat{\mathbf{V}}$

Kinetic Energy $\hat{\mathbf{T}} = -\frac{\hbar^2}{2\mu} \nabla^2$ where $\mu = \frac{m_e \cdot m_N}{m_e + m_N} \approx m_e$

Potential Energy $\hat{\mathbf{V}} = V(\mathbf{r}) = \frac{-Ze^2}{4\pi\epsilon_0 r}$ where $r = \sqrt{x^2 + y^2 + z^2}$

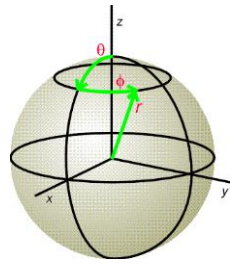
(electrostatic attraction between electron and nucleus).

$$\nabla^2 = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)$$



- We must solve the 2nd order differential equation:

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi + V(r)\psi = E\psi$$



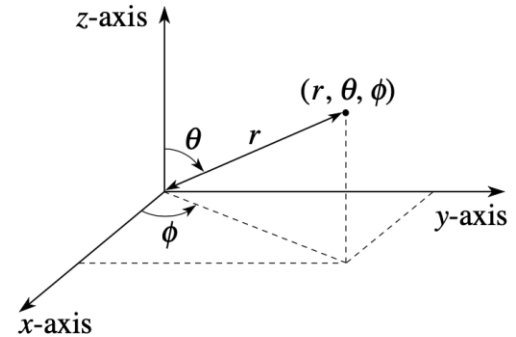
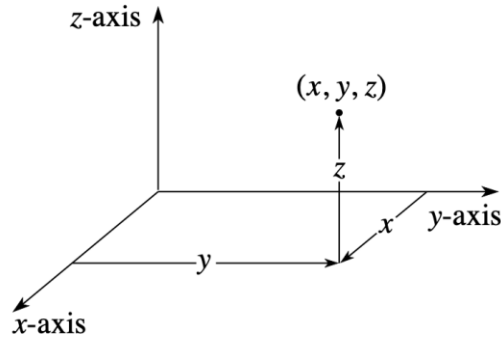
- Because of the spherical symmetry of the atom, it is convenient to describe the position of the electron in **spherical polar coordinates** (r, θ, ϕ) , rather than Cartesians (x, y, z) .

$$x = r \sin \theta \cos \phi$$

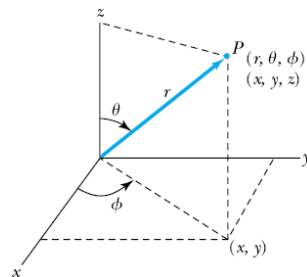
$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$r = \sqrt{x^2 + y^2 + z^2}$$



The Schrödinger Equation in Spherical Coordinates



$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{2\mu}{\hbar^2} (E - V) \psi = 0$$

- The wave function ψ is a function of r, θ, ϕ . This is a potentially complicated function.
- Assume instead that ψ is **separable**, that is, a product of three functions, each of one variable only:

$$\psi(r, \theta, \phi) = R(r) f(\theta) g(\phi)$$

$$\frac{\partial \psi}{\partial r} = fg \frac{\partial R}{\partial r} \quad \frac{\partial \psi}{\partial \theta} = Rg \frac{\partial f}{\partial \theta} \quad \frac{\partial^2 \psi}{\partial \phi^2} = Rf \frac{\partial^2 g}{\partial \phi^2}$$

Quantum Numbers

$$\Psi_{n,\ell,m_\ell}(r,\theta,\phi) = R_{n,\ell}(r) \cdot Y_{\ell,m_\ell}(\theta,\phi)$$

- ψ depends on 3 quantum numbers (n , ℓ , m_ℓ).

1. Principal Quantum Number (n)

- Positive integer $n = 1, 2, 3 \dots$
- For H and 1-e ions, the electron energy depends only on n

$$E_n = - \left(\frac{\mu e^4}{8h^2 \epsilon_0^2} \right) \cdot \left(\frac{Z^2}{n^2} \right)$$

2. Orbital Angular Momentum Quantum Number (ℓ)

- For a given value of n , ℓ can take the integer values:

$$\ell = 0, 1, 2 \dots (n-1)$$

e.g. $n = 1$ $\ell = 0$

$n = 2$ $\ell = 0, 1$

$n = 3$ $\ell = 0, 1, 2$

3. Orbital Magnetic Quantum Number (m_ℓ)

- For a given value of ℓ , m_ℓ can take the integer values:

$$m_\ell = 0, \pm 1, \pm 2 \dots \pm \ell$$

e.g. $\ell = 0$ $m_\ell = 0$

$\ell = 1$ $m_\ell = 0, \pm 1$

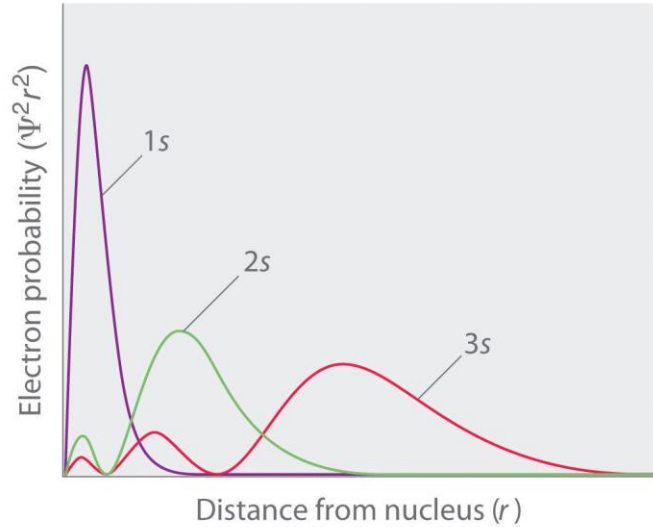
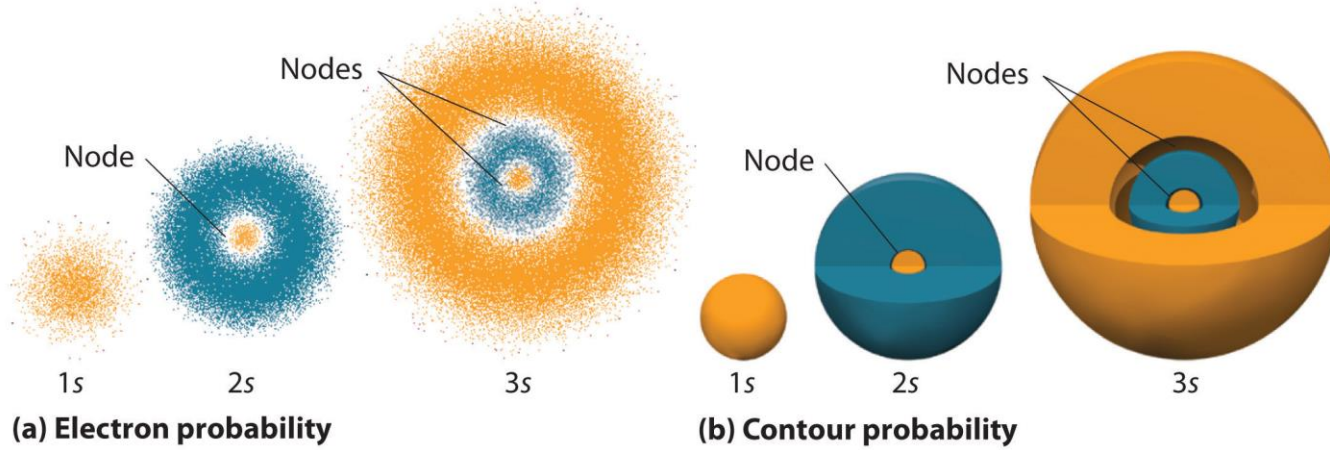
- There are $(2\ell+1)$ allowed values of m_ℓ .

Solutions of the Schrödinger Wave Equation for a One-Electron Atom

n	ℓ	m_ℓ	Orbital	Solution
1	0	0	$1s$	$\psi_{1s} = \frac{1}{\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} e^{-\sigma}$
2	0	0	$2s$	$\psi_{2s} = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} (2 - \sigma) e^{-\sigma/2}$
2	1	0	$2p_z$	$\psi_{2p_z} = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma e^{-\sigma/2} \cos \theta$
2	1	± 1	$\begin{cases} 2p_x \\ 2p_y \end{cases}$	$\psi_{2p_x} = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma e^{-\sigma/2} \sin \theta \cos \phi$ $\psi_{2p_y} = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma e^{-\sigma/2} \sin \theta \sin \phi$

Atomic Orbitals

- The wavefunctions (Ψ_{n,ℓ,m_ℓ}) describing an electron in an atom are known as **atomic orbitals**.
- Each atomic orbital (1-e wavefunction) is uniquely defined by 3 quantum numbers (n, ℓ, m_ℓ).
- The orbital (Ψ) gives the spatial distribution of the electron (via the Born interpretation of $|\Psi|^2$).
- Orbitals are often drawn as 3-D surfaces which enclose approx. 90% of the probability of finding the electron.



(c) Radial probability

Solutions of the Schrödinger Wave Equation for a One-Electron Atom

n	ℓ	m_ℓ	Orbital	Solution
1	0	0	1s	$\psi_{1s} = \frac{1}{\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} e^{-\sigma}$
2	0	0	2s	$\psi_{2s} = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} (2 - \sigma) e^{-\sigma/2}$
2	1	0	$2p_z$	$\psi_{2p_z} = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma e^{-\sigma/2} \cos \theta$
2	1	± 1	$2p_x$	$\psi_{2p_x} = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma e^{-\sigma/2} \sin \theta \cos \phi$
			$2p_y$	$\psi_{2p_y} = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma e^{-\sigma/2} \sin \theta \sin \phi$

Quantum Numbers and Atomic Orbitals

- **Orbitals** differ from each other in *size, shape and orientation*:
 - the orbital **size** is defined by the *principle quantum number* **n**
 - the type of orbital (it's **shape**) is defined by the *angular momentum quantum number* **ℓ**
 - the **orientation** of the orbital is defined by the *orbital magnetic quantum number* **m_ℓ**

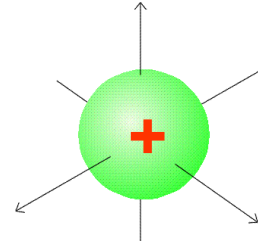
Shape of Atomic Orbitals

Shape – determined by the angular wavefunction $Y_{\ell,m\ell}(\theta,\phi)$

1s orbital ($n = 1, \ell = 0$)

- Normalized wavefunction:
- Has no angular dependence (spherically symmetric, depends only on r).
- All s orbitals are spherically symmetrical.

$$\psi(1s) = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0} \right)^{3/2} e^{-r/a_0}$$



2s orbital

$$\psi(2s) \propto (2 - r)e^{-r/2}$$

ns orbital

$$\psi(ns) \propto f(r)e^{-r/n}$$

- As $n \uparrow$, orbitals expand – average radius $\langle r \rangle \uparrow$

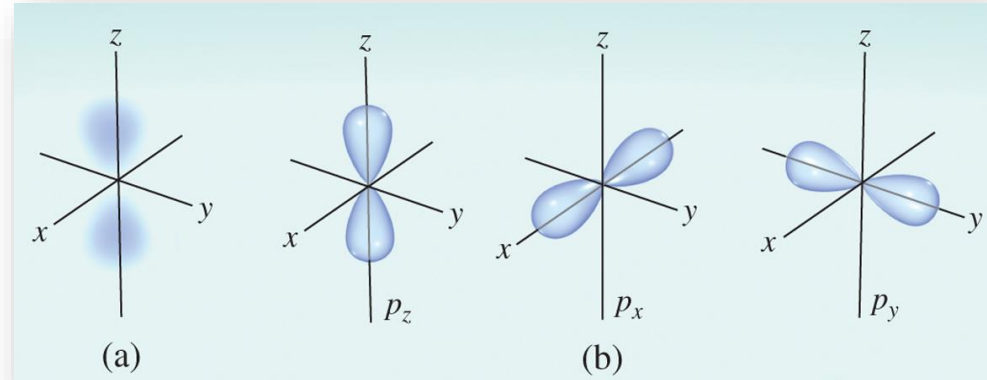
2p orbitals

$$(n = 2, \ell = 1)$$

- There are 3 degenerate orbitals, with 3 different m_ℓ values ($0, \pm 1$).
- e.g. $m_\ell = 0 \Rightarrow 2p_z$ orbital (pointing along z-axis)

$$\psi(2p_z) = \frac{1}{4\sqrt{2\pi}} \left(\frac{1}{a_0} \right)^{5/2} \cdot r \cos\theta \cdot e^{-r/2a_0}$$

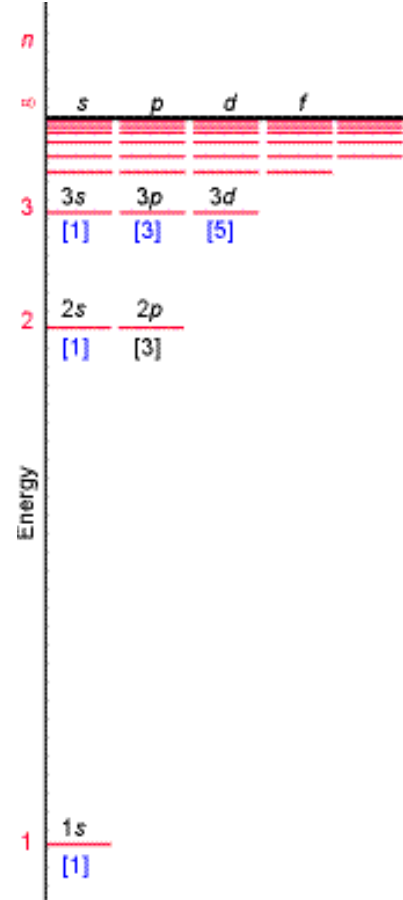
- $m_\ell = \pm 1 \Rightarrow 2p_x, 2p_y$ orbitals



Three orientations:
 $\ell = 1$ (as required for a p orbital)
 $m_\ell = -1, 0, +1$

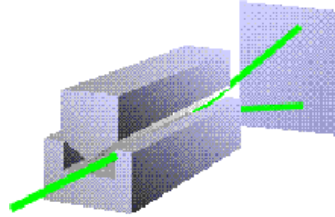
Atomic Orbital Energy Diagram for H

- Orbital energies depend only on the principal quantum number n .
- For a given n value (shell), all sub-shells (ℓ) and orbitals (m_ℓ) have the same energy
 - i.e. they are **degenerate**.



Electron Spin

- Stern and Gerlach (1921) observed that a beam of Ag atoms is split into 2 beams by an inhomogeneous magnetic field.



- Spin angular momentum (**S**) has magnitude:

$$|S| = \sqrt{s(s+1)}\hbar = \frac{\sqrt{3}\hbar}{2}$$

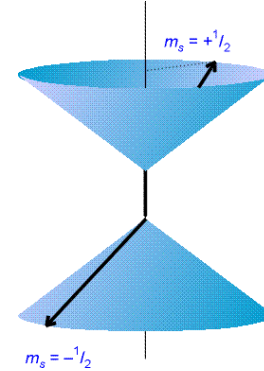
- The projection of **S** on the z-axis:

$$S_z = m_s \hbar = \pm \frac{1}{2} \hbar$$

- There are two possible electron spin states:

$$\begin{array}{ll} m_s = +\frac{1}{2} & \text{“spin up” } \uparrow \\ m_s = -\frac{1}{2} & \text{“spin down” } \downarrow \end{array}$$

- Spin is an intrinsic property of the electron and is not connected with orbital motion.
- Complete specification of an electron in an atom requires 4 quantum numbers (n, ℓ, m_ℓ, m_s) – as **s** is fixed.



Quantum Numbers (QN)

Principal QN:

$n = 1, 2, 3, 4, \dots$

Angular momentum QN:

$l = 0, 1, 2, 3, \dots (n - 1)$

Rule: $l = (n - 1)$

Magnetic QN:

$m_l = \dots -2, -1, 0, 1, 2, \dots$

Rule: $-l, \dots, 0, \dots, +l$

TABLE 1.3 Quantum Numbers for Electrons in Atoms

Name	Symbol	Values	Specifies	Indicates
principal	n	$1, 2, \dots$	shell	size
orbital angular momentum*	l	$0, 1, \dots, n - 1$	subshell: $l = 0, 1, 2, 3, 4, \dots$ $s, p, d, f, g, \dots$	shape
magnetic	m_l	$l, l - 1, \dots, -l$	orbitals of subshell	orientation
spin magnetic	m_s	$+\frac{1}{2}, -\frac{1}{2}$	spin state	spin direction

*Also called the *azimuthal quantum number*.